

MEDS Module 4 Section 8

Name: Umer Faisal

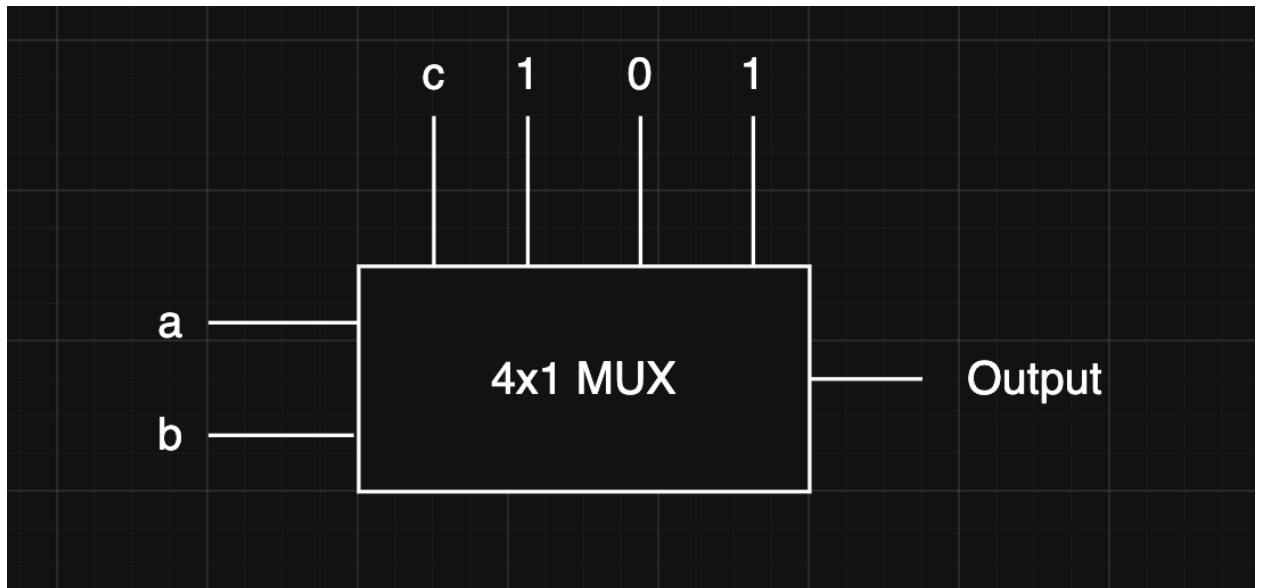
TASK 8: $F = \sum m(1,2,3,6,7)$ Using a 4x1 MUX

1. EDA Playground Link: <https://edaplayground.com/x/u4gF>

2. Truth Table:

000	0
001	1
010	1
011	1
100	0
101	0
110	1
111	1

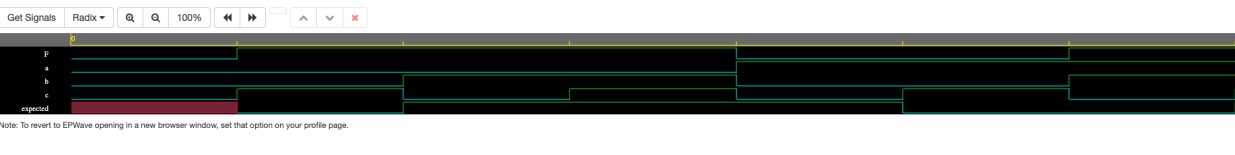
3. Schematic Diagram:



4. Terminal Output Screenshot:

```
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 25 17:32 2026
a b c | F | Expected
-----
0 0 0 | 0 | 0
0 0 1 | 1 | 1
0 1 0 | 1 | 1
0 1 1 | 1 | 1
1 0 0 | 0 | 0
1 0 1 | 0 | 0
1 1 0 | 1 | 1
1 1 1 | 1 | 1
Testing complete.
```

5. Waveform Screenshot:



TASK 9: 4-bit Barrel Shifter (Rotate) Using unique case

1. EDA Playground Link: <https://edaplayground.com/x/ubTd>

2. Terminal Output Screenshot:

```
Log Share
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 25 17:56 2026
W shift | Y k
-----
0000 0 | 0000 0
0000 1 | 0000 0
0001 0 | 0001 0
0001 1 | 0000 1
0010 0 | 0010 0
0010 1 | 0001 0
0011 0 | 0011 0
0011 1 | 0001 1
0100 0 | 0100 0
0100 1 | 0010 0
0101 0 | 0101 0
0101 1 | 0010 1
0110 0 | 0110 0
0110 1 | 0011 0
0111 0 | 0111 0
0111 1 | 0011 1
1000 0 | 1000 0
1000 1 | 0100 0
1001 0 | 1001 0
1001 1 | 0100 1
1010 0 | 1010 0
1010 1 | 0101 0
1011 0 | 1011 0
1011 1 | 0101 1
1100 0 | 1100 0
1100 1 | 0110 0
1101 0 | 1101 0
1101 1 | 0110 1
1110 0 | 1110 0
1110 1 | 0111 0
1111 0 | 1111 0
1111 1 | 0111 1
$finish called from file "design.sv", line 39.
$finish at simulation time 32
VCS Simulation Report
Time: 32 ns
```

3. Waveform Screenshot:

